Jean-Guillaume de Damas

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Experience

CERMICS, CNRS, ENPC, Institut Polytechnique de Paris, Postdoctoral Researcher — Marne-la-Vallée, France
May 2026 – present
7 months

- Working at the Centre d'Enseignement et de Recherche en Mathématiques et Calcul Scientifique on the development of hybrid model reduction methods with Virginie Ehrlacher
- Developing numerical methods for accelerating parametric studies in granular material simulations
- Implementing algorithms in the software SCOPI++

CERFACS, Master Intern - Researcher & Scientific Programmer — Toulouse, France
Apr 2022 – Oct 2022
7 months

- Researched randomized SVD preconditioners with applications to data assimilation
- Delivered functional C++ code for the European Weather Forecast Center
- Supervised by Selime Gürol

Education

PhD **INRIA Paris & Sorbonne Université**, Applied Mathematics — Paris, France
Nov 2022 – Mar 2026

- Thesis: Randomized methods for solving eigenvalue problems and computing low-rank matrix approximations
- Obtained up to 3-4x speed-up compared to state of the art methods while maintaining accuracy in the computations
- Supervised by Laura Grigori within the ERC project EMC2

MSc **ISAE-SUPAERO**, Engineering — Toulouse, France
Sept 2018 – Nov 2022

- Machine learning: Algorithms for SVM, Neural Networks, Reinforcement learning and mathematical background
- Statistics: Bayesian inference, Stochastic algorithms, Non-parametric estimation
- Fluid dynamics: Theory, computational aspects, wind tunnel demonstration
- High performance computing: Parallelization in C
- Optimisation: Inverse problem, data assimilation, problem regularization, and numerical challenges in inexact arithmetic

Publications

Randomized methods for solving eigenvalue problems and computing low-rank matrix approximations — June 2026
Jean-Guillaume de Damas
10.70675/42a47550z0087z4fddzbf44z13273ef37f8b (PhD Thesis)

Randomized Implicitly Restarted Arnoldi Method for the Non-Symmetric Eigenvalue Problem — Oct 2025
Vol. 46, No. 4, pp. 2395–2422. ISSN: 1095-7162.
Jean-Guillaume de Damas, Laura Grigori
10.1137/24m1674352 (SIAM Journal on Matrix Analysis and Applications)

Randomized Krylov-Schur Eigensolver with Deflation — Aug 2025
Jean-Guillaume de Damas, Laura Grigori
10.48550/ARXIV.2508.05400 (arXiv)

Randomized Orthogonalization and Krylov Subspace Methods: Principles and Algorithms Nov 2025
Jean-Guillaume de Damas, Laura Grigori, Igor Simunec, Edouard Timsit
10.48550/arXiv.2512.15455 (arXiv)

Selected Talks

Workshop on Approximate Computing in Numerical Linear Algebra Oct 2025
A randomized Krylov-Schur eigensolver, featuring mixed-precision and deflation

CERMICS Applied Mathematics Seminar June 2025
Randomized methods for the eigenvalue problem and low rank approximation

SIAM Conference on Applied Linear Algebra LA24 & CERFACS Sparse Days May 2024
Randomized Implicitly Restarted Arnoldi algorithm for the unsymmetric eigenvalue problem

Teaching

Analysis and Partial Differential Equations Part II (24h) Spring 2027
Ecole nationale des ponts et chaussées (ENPC), Champs-sur-Marne

Analysis and Partial Differential Equations Part I (24h) Winter 2026
Ecole nationale des ponts et chaussées (ENPC), Champs-sur-Marne

Mathematics for Science Studies I (52h) Fall 2023
Sorbonne University, Paris

Programming Basics I (20h) Fall 2023
Sorbonne University, Paris

Skills

Programming
Julia, Python, Matlab, R, C, C++

Technologies
PyTorch, Scikit-learn, OpenMP, MPI, Cuda

Expertise
Numerical Analysis, Randomized Linear Algebra, Eigensolvers, Low-Rank Approximations

Languages

English
Advanced Level C1 (TOEFL ITP 600/677)

German
Intermediate (A2)

Japanese
Beginner (A1)

Outside of the office

- Piano & Musicology: Learning as a passionate amateur with conservatory-level classes
- Game developement: Building its own engine in C++ or using Unity/Unreal